

Week 4 Assignment: Planning in a Stochastic Gridworld

MDP modeling, value iteration, policy iteration, and verification

CMPSC 480 - Instructor Solution Deck

Instructor material - do not distribute with the question deck

Solution overview

This instructor deck provides a complete matched solution and grading guidance.

Coverage: Week 4: MDPs, Bellman backups, value iteration, and policy iteration

Points

10 total

Expected time

6-8 hours

Submission

Repository plus experiment memo

Instructor verification checklist

- ▶ Use Python 3.11 and NumPy only unless the prompt explicitly permits another package.
- ▶ Preserve the published interfaces and make all tie-breaking deterministic.
- ▶ State assumptions, validate inputs, and handle empty, terminal, and unreachable cases.
- ▶ Include focused tests whose names explain the defect each test would expose.
- ▶ Report measured evidence; do not replace experimental results with unsupported claims.

Solution 1.a - MDP model (1 points)

Question focus. Define S , A , P , R , γ , initial state, walls, and terminal states.

Answer. States are legal grid cells; actions are cardinal moves; intended motion has probability 0.8 and perpendicular slips 0.1 each. Blocked motion stays in place. Rewards and terminal semantics are stated, with γ in $[0,1)$.

Solution 1.b - MDP model (1 points)

Question focus. Write validation tests for every state-action transition distribution.

Answer. For each legal nonterminal pair, successor probabilities are nonnegative and sum to one within tolerance; all successors are legal states. Terminal states expose no actions or a documented absorbing convention.

Solution 2.a - Value iteration (1 points)

Question focus. Write the Bellman optimality backup used for each nonterminal state.

Answer. $V_{\text{new}}(s)$ is the maximum over actions of the sum over successors of $P(s\text{-prime}|s,a)$ times $[R(s,a,s\text{-prime}) + \gamma V_{\text{old}}(s\text{-prime})]$. Terminal values follow the declared reward convention.

Solution 2.b - Value iteration (1 points)

Question focus. Use a copy of the previous value vector and stop when the maximum residual is below tolerance.

Answer. Synchronous updates read only V_{old} . The residual is $\max_s \text{absolute}(V_{new}(s) - V_{old}(s))$; iteration count has a safety bound and tolerance is positive.

Solution 2.c - Value iteration (1 points)

Question focus. Extract a deterministic greedy policy and report values for designated cells.

Answer. Each action is scored with the final Bellman lookahead; stable action order resolves ties. Reported cells match an independently computed oracle within tolerance.

Solution 3.a - Policy iteration (1 points)

Question focus. Describe iterative policy evaluation for a fixed policy.

Answer. Repeatedly apply the Bellman expectation backup using only the current policy action until the evaluation residual is below tolerance.

Solution 3.b - Policy iteration (1 points)

Question focus. Implement greedy improvement and state the stability condition.

Answer. For every nonterminal state choose an action maximizing one-step lookahead under the evaluated values. Stop when no chosen action changes under deterministic ties.

Solution 3.c - Policy iteration (1 points)

Question focus. Compare the final policy and values with value iteration.

Answer. Optimal values agree within a declared numerical tolerance, and policies are equal or choose actions with equal optimal action value.

Solution 4.a - Validation and analysis (1 points)

Question focus. Test $\gamma=0$, a one-state terminal MDP, blocked slips, and a reward perturbation.

Answer. At $\gamma=0$ only immediate expected reward matters. The terminal case is stable. Blocked slips accumulate probability on staying put. Increasing one terminal reward should not reduce optimal value in reachable states.

Solution 4.b - Validation and analysis (1 points)

Question focus. Explain how gamma and slip probability change the policy in two experiments.

Answer. The answer compares controlled runs, reports values and actions, and links larger gamma to future rewards while larger slip can favor safer routes.

Edge-case reasoning and expected behavior

- ▶ Return the terminal convention immediately.
- ▶ Merge that probability into the stay-put successor.
- ▶ Use documented stable action order.
- ▶ Backups reduce to immediate expected reward.
- ▶ Reject the configuration before iteration.

Grading rubric

Category	Points	Full-credit evidence
MDP model	2	Probabilities normalize and terminal/wall semantics are explicit.
Value iteration	3	Backups, convergence, and terminal handling are correct.
Policy iteration	3	Evaluation/improvement loops and stopping rule are sound.
Validation and analysis	2	Tests expose modeling errors and conclusions follow evidence.

Reflection exemplar

Answer. A strong answer isolates a transition, reward, terminal, gamma, or slip convention, reports before-and-after values or policies, and explains why both solvers changed consistently.

Grading guidance. Award credit for a specific claim supported by technical evidence from the submitted work.